Web GIS Lab 02

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District Explorer Interactive Web Map of Pakistan

Item	Details
Student	Muhammad Ahmad
Roll No.	538160
Programme	IGIS/SCEE, NUST
Lab	Web GIS Lab 02 — District Explorer
Tools	QGIS, VS Code, Leaflet.js, GitHub Pages

1. Published Links

- **GitHub repository:** [Insert repository URL]
- **Live map (GitHub Pages):** [Insert GitHub Pages URL]

Note: yellow-highlighted text marks values that were not visible in the screenshots and must be filled in from your own measurements before submission.

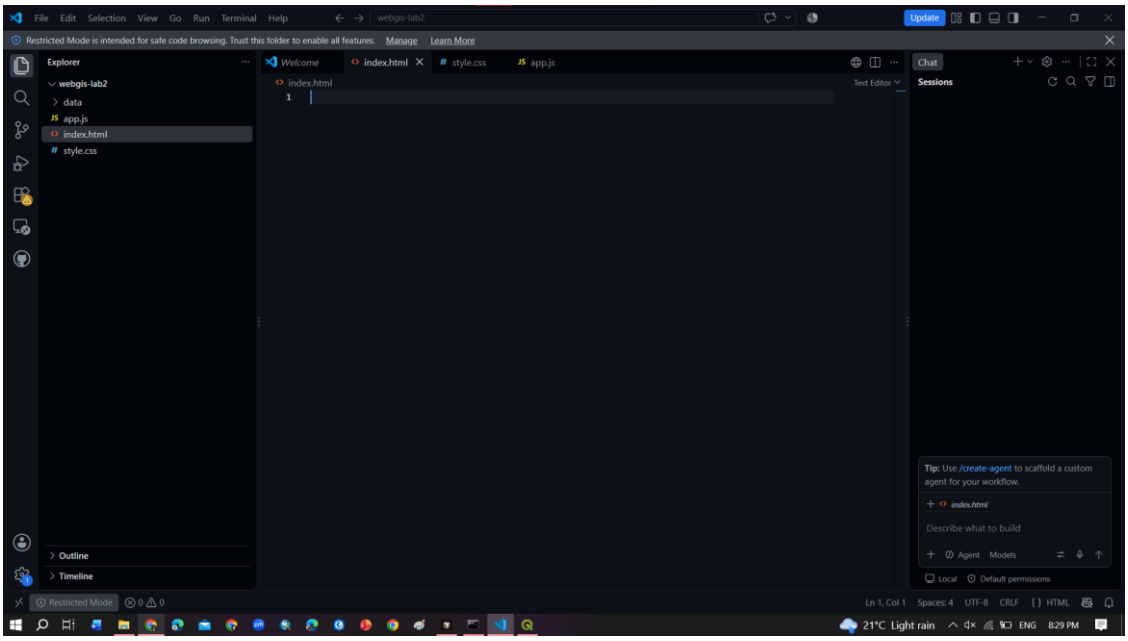
2. Overview

This lab builds an interactive web map of Pakistan's 160 districts. District polygons were prepared in QGIS, exported to GeoJSON (four versions with different precision and simplification), and displayed with Leaflet.js on top of an OpenStreetMap basemap. Districts are shaded by 2023 census population in five classes, with a separate grey class for missing data. A province dropdown filters the map, and clicking a district opens a popup with its attributes. The final site is published through GitHub Pages. The sections below document each of the nine checkpoints.

3. Checkpoint 1 — Project Setup

The project folder was created and opened in VS Code. It contains the three required files (index.html, style.css, app.js) and an empty data folder. At this stage the files are still empty (Line 1, Col 1 in index.html).

Item	Observed
Project folder	webgis-lab2
Files	index.html, style.css, app.js
Sub-folder	data (empty)



Checkpoint 1: VS Code showing the project folder, three files and the data folder.

4. Checkpoint 2 — GeoJSON Versions and File Sizes

The district shapefile was inspected in QGIS and exported to four GeoJSON versions (A: straight conversion; B: coordinate precision 5; C: simplified with 0.001° tolerance (~100 m) and precision 5; D: simplified with 0.01° tolerance (~1 km) and precision 5). Both C and D were generated from the original layer. Version C was copied into the project as data/districts.geojson.

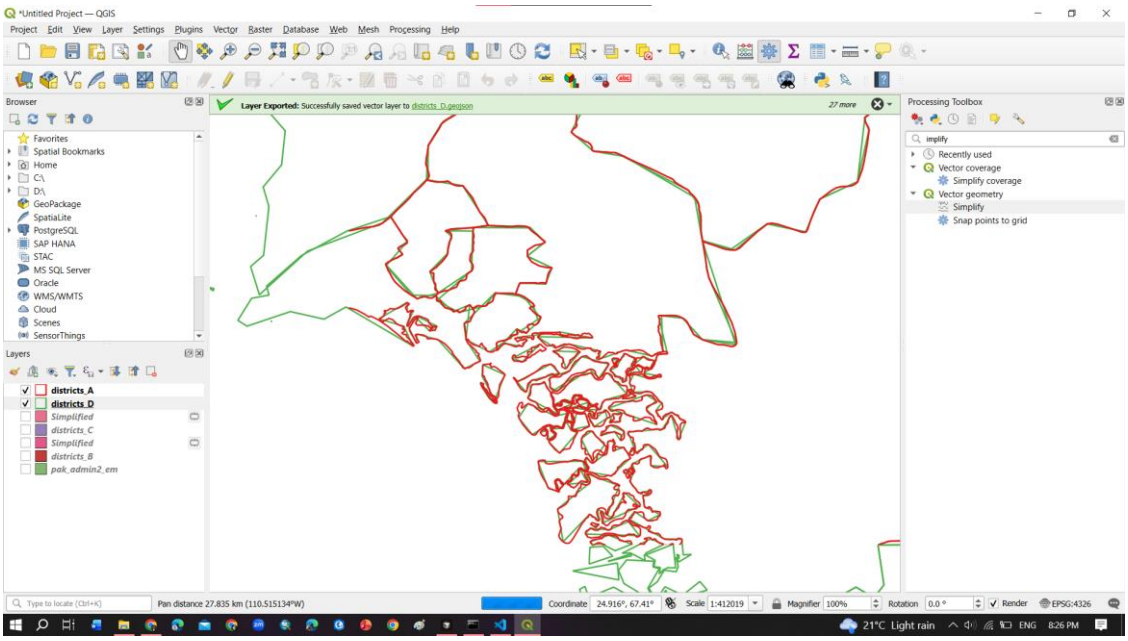
4.1 File-size measurement table

Version	Method	Size	% of A
Shapefile	Sum of five adm2 components	4376000	N/A
A	Straight conversion	10146902	100%
B	Precision 5	5139791	50.56%
C	Simplify 100 m, precision 5	980883	9.67 %
D	Simplify 1 km, precision 5	285341	2.81%

Formula: % of A = (Version size ÷ A size) × 100. Sizes are taken from File Explorer → Properties.

4.2 Detail loss: Version A vs Version D

The screenshot overlays districts_A (red outline) and districts_D (green outline) at a large scale (1:412,019). Version A follows the boundaries closely with many small vertices, while Version D replaces fine wiggles with long straight segments. The 1 km simplification is therefore visibly coarser, and in the south of the view it also produces sharp angular shapes. This is why D is unsuitable for detailed district-level analysis.



Checkpoint 2: districts_A (red) and districts_D (green) drawn together in QGIS.

5. Checkpoint 3 — GeoJSON Inspection

The GeoJSON file was opened in VS Code and its structure was inspected. The answers are summarised below.

Question	Answer
1. Outermost object type	FeatureCollection
2. Number of features	160 (confirmed in the browser Console, Checkpoint 5)
3. First feature geometry type	MultiPolygon in the inspected file (the loaded districts.geojson reports Polygon in the Console)
4. Nested bracket levels before first number	4 levels for MultiPolygon: [[[[73.53892, 34.12617] ...
5. Keys in first feature properties	adm2_name, adm2_name1–3, adm2_pcode, adm1_name, adm1_name1–3, adm1_pcode, adm0_name, adm0_name1–3, adm0_pcode, valid_on, valid_to, area_sqkm, version, lang, lang1–3, center_lat, center_lon
6. First coordinate pair	[73.53892, 34.12617] (longitude, latitude)
7. Coordinate order check	Correct: longitude 73.5 lies within 61–77 and latitude 34.1 within 24–37, matching the GeoJSON lon, lat order
8. Polygon ring check	The first and last coordinate pairs of the outer ring must be identical (closed ring), as required by the GeoJSON specification

Note: the property keys above are from the raw source file. The final map file (Checkpoint 5) uses the simplified attribute set district, province, pop_2023, area_km2 and pcode.

1. Outermost object `type`: **`FeatureCollection`**

2. Number of features in the `features` array: The file contains multiple district features. The exact total needs to be counted from the complete file.

3. First feature's geometry type: `MultiPolygon`

4. Number of nested square-bracket levels before the first coordinate number: 4 levels. The coordinate structure begins with `[[[[73.53892, 34.12617]...`

5. Keys in the first feature's `properties` object:

- `adm2_name`
- `adm2_name1`
- `adm2_name2`
- `adm2_name3`
- `adm2_pcode`
- `adm1_name`
- `adm1_name1`
- `adm1_name2`
- `adm1_name3`
- `adm1_pcode`
- `adm0_name`
- `adm0_name1`
- `adm0_name2`
- `adm0_name3`
- `adm0_pcode`
- `valid_on`
- `valid_to`
- `area_sqkm`
- `version`
- `lang`
- `lang1`
- `lang2`
- `lang3`
- `center_lat`
- `center_lon`

Coordinate Order Check

Find the first coordinate pair. For Pakistan, the approximate ranges are:

- Longitude: 61–77
- Latitude: 24–37

In GeoJSON, the coordinate order is longitude, latitude. Therefore, the first number should usually be in the 60s or 70s, and the second number should be in the 20s or 30s.

Polygon Ring Check

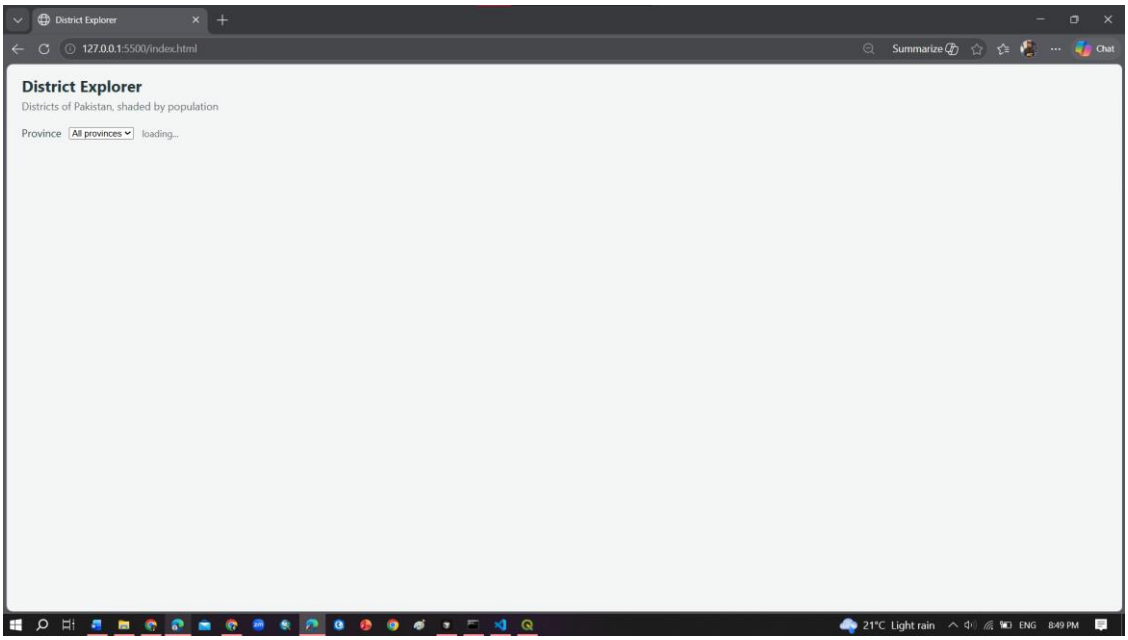
For the outer ring of the first polygon:

- Look at the first coordinate pair.
- Look at the last coordinate pair.
- Both pairs should be identical.

Checkpoint 3: GeoJSON inspection notes.

6. Checkpoint 4 — Empty Page Test with Live Server

index.html and style.css were tested before adding any JavaScript. The page shows the header (“District Explorer”), the subtitle, the Province dropdown with “All provinces”, and the status text “loading...”. The map area is empty because app.js had not yet been added. The URL (127.0.0.1:5500/index.html) confirms the page was served by Live Server over HTTP rather than opened as file:///.



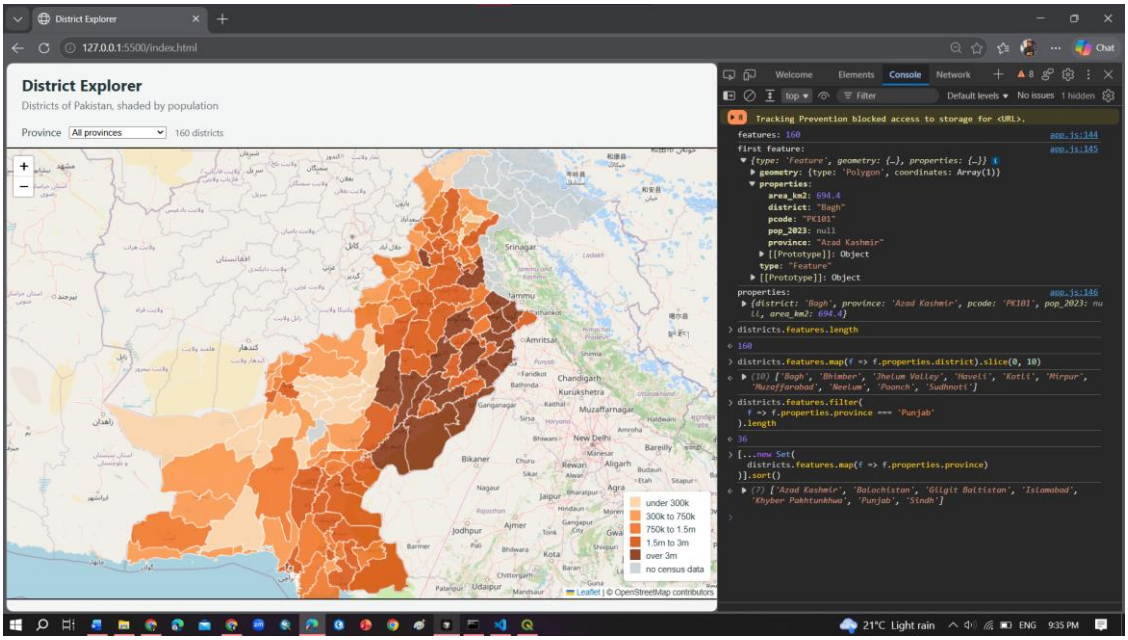
Checkpoint 4: header, empty map area and the 127.0.0.1 URL.

7. Checkpoint 5 — Console Verification

After adding app.js, the map loaded and the Console logs were checked. The results are:

Console check	Result
districts.features.length	160
First feature	Type Feature, geometry Polygon; district Bagh, province Azad Kashmir, pcode PK101, pop_2023 null, area_km2 694.4
First ten districts	Bagh, Bhimber, Jhelum Valley, Haveli, Kotli, Mirpur, Muzaffarabad, Neelum, Poonch, Sudhnoti
Punjab districts	36
Distinct provinces (7)	Azad Kashmir, Balochistan, Gilgit Baltistan, Islamabad, Khyber Pakhtunkhwa, Punjab, Sindh

The first feature (Bagh) has pop_2023 = null, which is consistent with Azad Kashmir not being covered by the 2023 census.



Checkpoint 5: Console output showing feature count, first feature and provinces.

8. Checkpoint 6 — Network Measurements

The Network tab was used with “Disable cache” enabled. The GeoJSON file is requested once, whereas the basemap is delivered as many separate 256 × 256 px PNG tiles that depend on the zoom level and visible extent.

8.1 Normal connection

- **districts_C.geojson (loaded as the map data):** 933 kB transferred, 1,367 kB resource size; load time 63 ms in the request list (54.13 ms in the Timing tab).
- **Page totals:** 28 requests, 1.4 MB transferred, Finish 1.14 s, DOMContentLoaded 471 ms, Load 1.10 s (one reload); 49 requests and 1.8 MB transferred in another reload with more tiles.
- **Map tile:** selected tile preview is 256 × 256 px, PNG, 19.1 kB. Tile requests are named 11.png–14.png in the list; tile URL pattern: tile.openstreetmap.org/{z}/{x}/{y}.png.

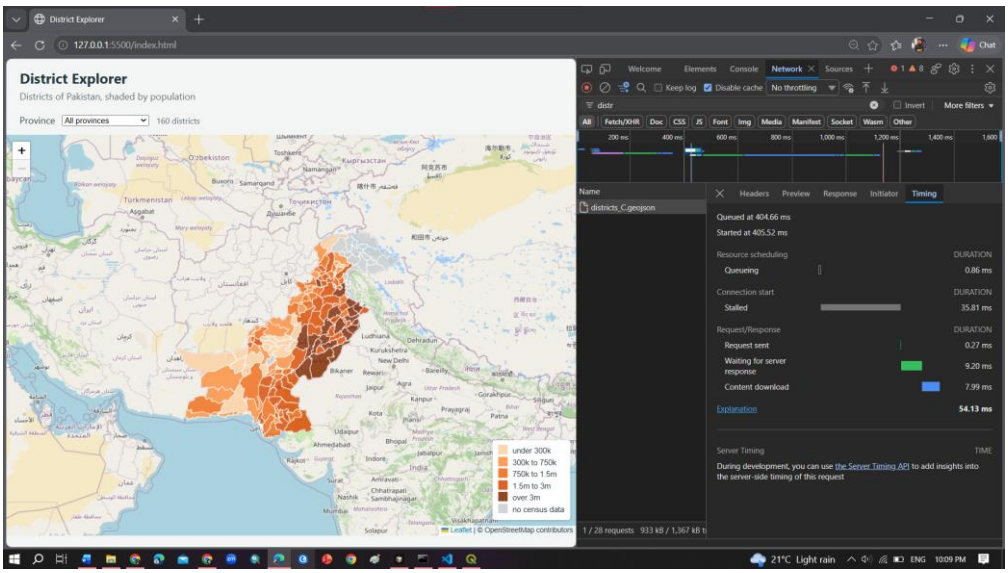
8.2 Slow 3G

- **districts_C.geojson:** load time 6.97 s (list shows 11.6 kB for the request); page finish 12.30 s, DOMContentLoaded 5.25 s, 27 requests.

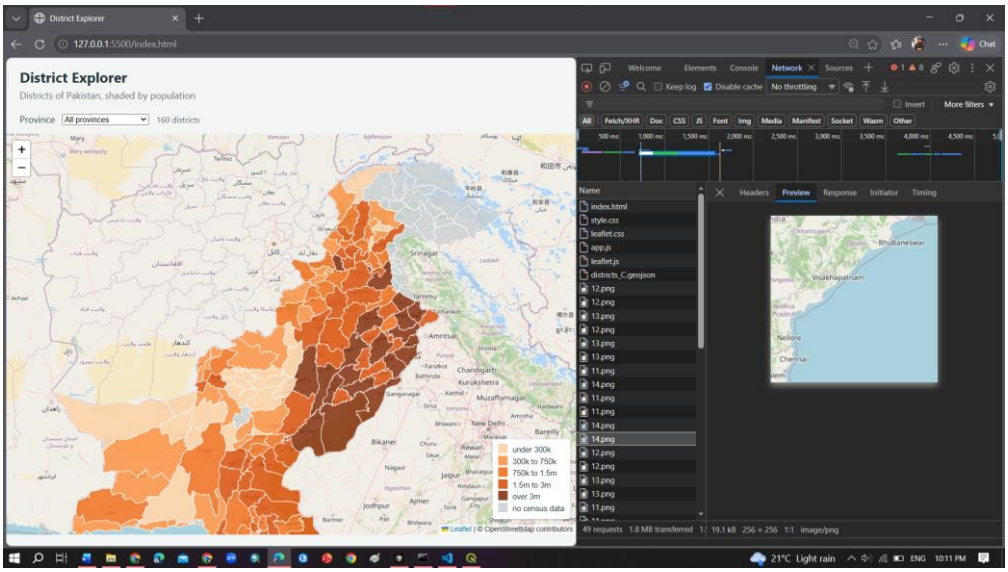
8.3 Summary table

Measurement	Normal connection	Slow 3G
districts.geojson size	933 kB	11.6 kB (as shown by browser)
districts.geojson load time	63 ms	6.97 s
districts_A.geojson size	___	___
districts_A.geojson load time	___	___
Number of PNG tile requests	___ (filter ".png")	—
Tile size / zoom level (z)	256 × 256 px	z = ___ (see Headers)

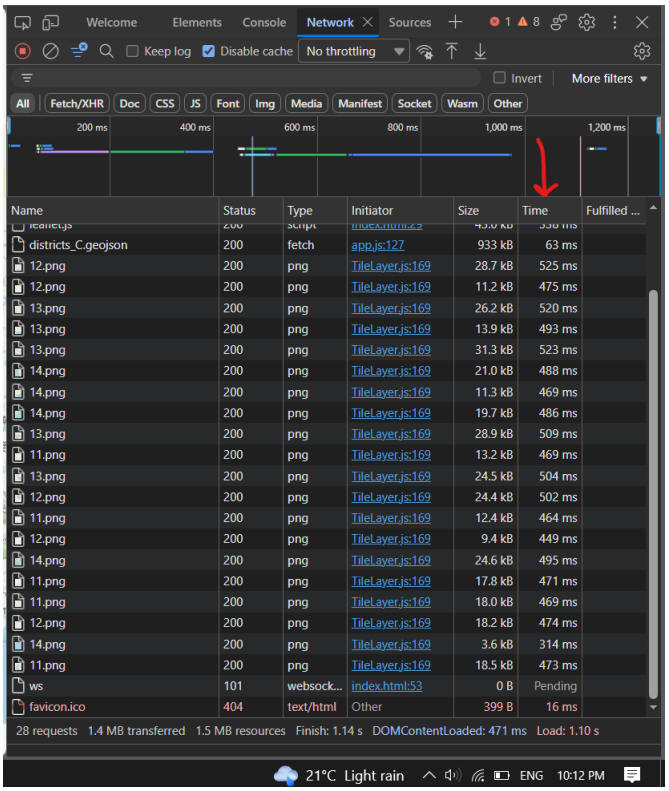
Observation: on Slow 3G the GeoJSON took roughly 100 times longer to load than on the normal connection, and every tile also took about 6–7 s. This shows why file size matters for web maps, and why the simplified Version C is preferable to Version A.



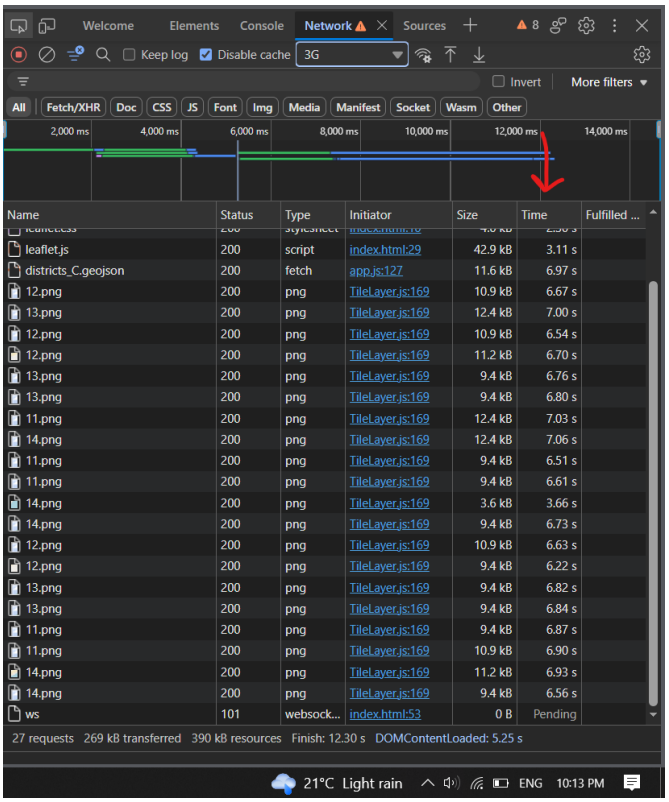
Checkpoint 6A: Timing tab for the GeoJSON request (54.13 ms).



Checkpoint 6B: Selected map tile preview (256 × 256 px, 19.1 kB).



Checkpoint 6C: Normal connection — GeoJSON 933 kB, 63 ms.



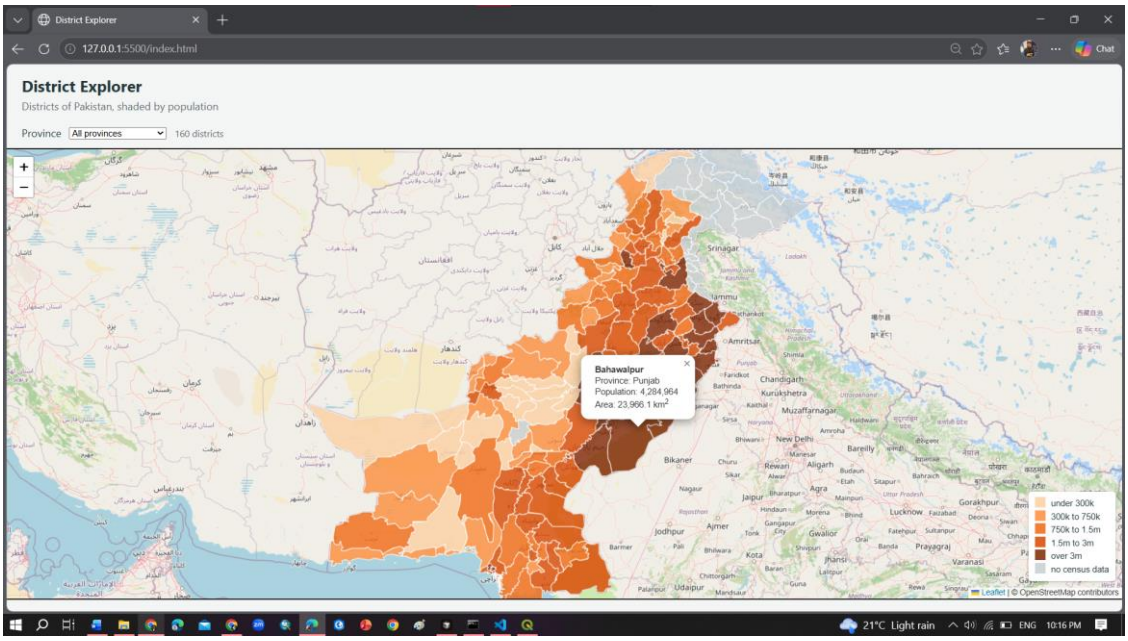
Checkpoint 6D: Slow 3G — GeoJSON load time 6.97 s.

9. Checkpoint 7 — Population Classification and Popup

Districts are shaded using five manual population classes plus a separate grey class for missing census data (Azad Kashmir, Gilgit-Baltistan and Lehri). The legend in the bottom-right corner uses the same colours as the map. Clicking Bahawalpur opens a popup with all four attributes: name, province, population and area.

Population	Colour	Hex
Above 3,000,000	Darkest brown	#7f2704
Above 1,500,000	Dark orange	#d94801
Above 750,000	Orange	#f16913
Above 300,000	Light orange	#fd8d3c
300,000 or below	Pale peach	#fdd0a2
Missing value (no census data)	Grey	#c9cfd1

Example popup: Bahawalpur — Province: Punjab; Population: 4,284,964; Area: 23,966.1 km². This district falls in the highest class (over 3 million) and appears in the darkest brown. Population ranges were divided into five classes using manual break values, and missing data was given its own grey class so it is not confused with low population.

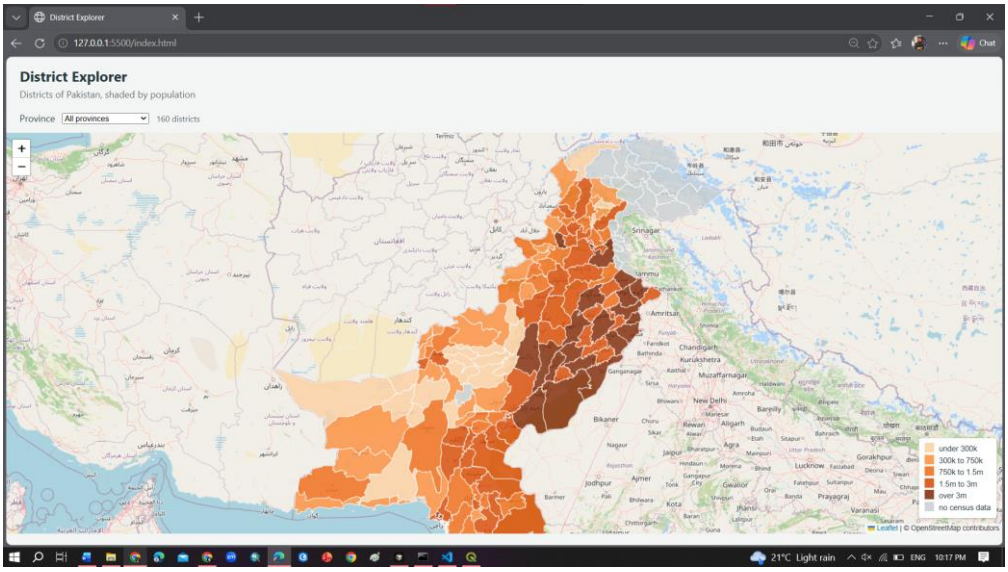


Checkpoint 7: coloured map, legend and an open popup for Bahawalpur.

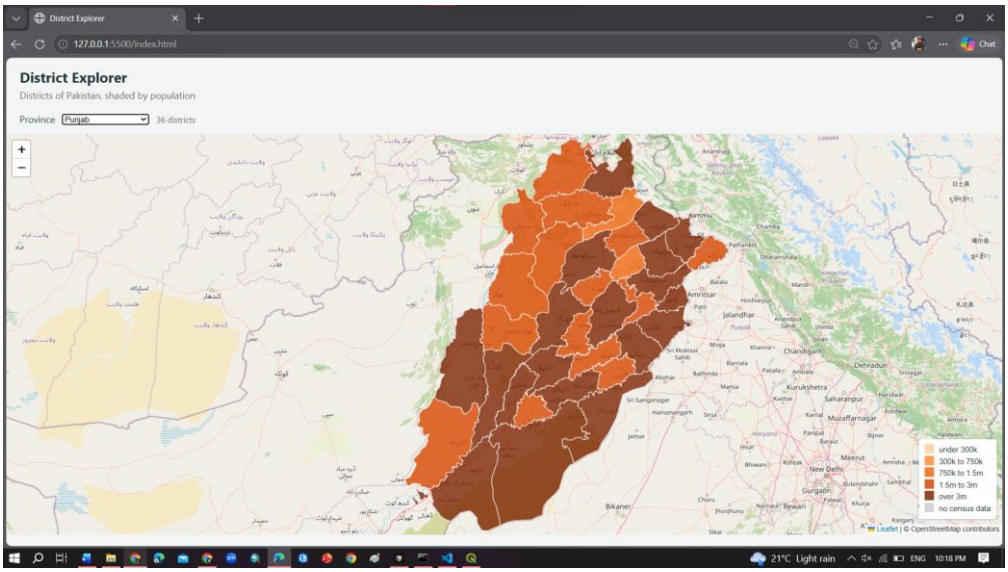
10. Checkpoint 8 — Province Filter

The province dropdown redraws the map with only the selected province's districts, zooms to their bounds, and updates the district count. Three states were tested:

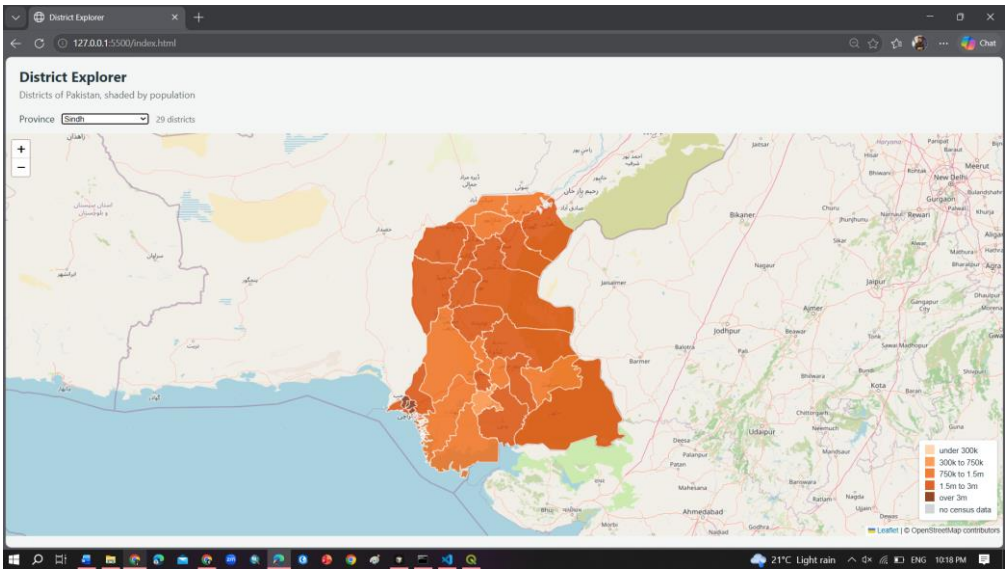
Selection	Districts	Result
All provinces	160	Entire country displayed
Punjab	36	Only Punjab districts displayed; map zooms to Punjab
Sindh	29	Only Sindh districts displayed; map zooms to Sindh



Checkpoint 8A: All provinces selected (160 districts).



Checkpoint 8B: Punjab selected (36 districts).



Checkpoint 8C: Sindh selected (29 districts).

11. Checkpoint 9 — Coordinate System Comparison

Areas were calculated in QGIS with the expression $area / 1000000 on two exported layers: d_3857 (EPSG:3857, Web Mercator) and d_32643 (EPSG:32643, WGS 84 / UTM zone 43N). Five districts spread from south to north were chosen: Awaran, Attock, Bagh, Abbottabad and Astore. The UTM area is the reference.

11.1 Measured UTM areas

District	Province / Region	Calculated area (km²)	Comparison
Bagh	Azad Kashmir	694.417	Smallest of the five
Abbottabad	Khyber Pakhtunkhwa	1,639.517	~2.4 × Bagh
Astore	Gilgit-Baltistan	5,132.454	Moderate mountainous area

District	Province / Region	Calculated area (km²)	Comparison
Attock	Punjab	6,910.050	~10 × Bagh
Awaran	Balochistan	24,989.609	Largest (large arid district)

11.2 Five-district comparison table

District	Latitude (approx.)	3857 area (km²)	UTM area (km²)	% diff. (measured)	Inflation factor $1/\cos^2\phi$	Predicted % diff.
Awaran	26.45°N	_____	24,989.609	_____	1.247	24.7%
Attock	33.77°N	_____	6,910.050	_____	1.447	44.7%
Bagh	33.98°N	_____	694.417	_____	1.454	45.4%
Abbottabad	34.15°N	_____	1,639.517	_____	1.460	46.0%
Astore	35.33°N	_____	5,132.454	_____	1.502	50.2%

The UTM areas come from the screenshots. The 3857 areas were not in the screenshots, so fill them in from the d_3857 layer and calculate % difference = $(A_{3857} - A_{UTM}) \div A_{UTM} \times 100$. Latitudes are approximate district-centre values, so replace them with your own readings if you have them. The inflation factors and predicted differences are calculated from those latitudes.

11.3 Answers

Q1. Does the percentage difference grow or shrink as you move north? It grows. Web Mercator stretches distances in both directions by $1/\cos\phi$, so areas are inflated by about $1/\cos^2\phi$. This factor increases with latitude, from about 1.25 at Awaran (~26.5°N) to about 1.50 at Astore (~35.3°N). The northern districts therefore show the largest percentage difference.

Q2. Predicted inflation factor. Inflation factor = $1 \div \cos^2(\phi)$, with ϕ in degrees. The factors for each district are in the table above; for example, at 35.33°N the factor is about 1.50, meaning the Web Mercator area is predicted to be about 50% larger than the true area.

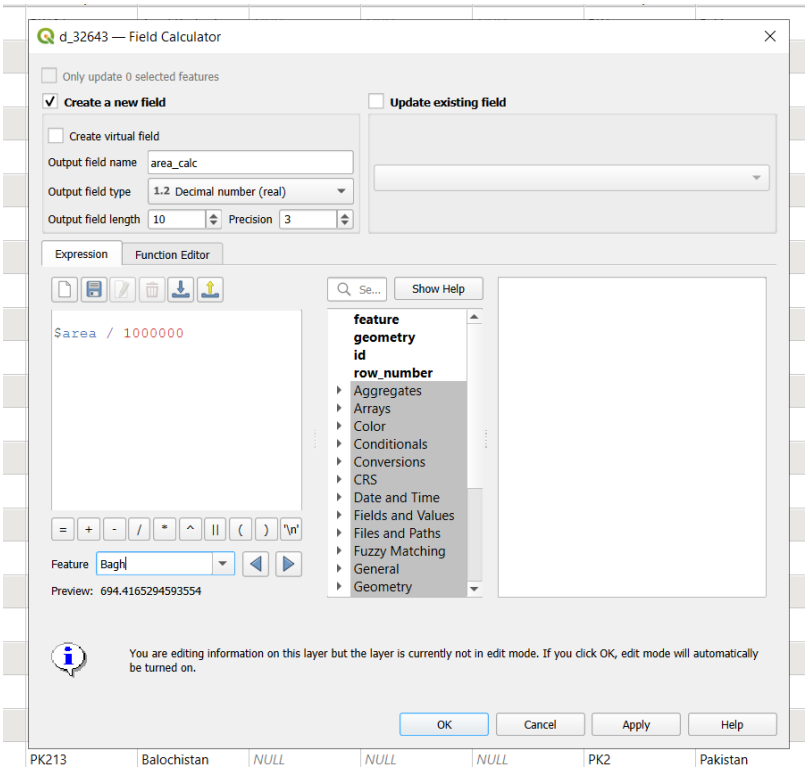
Q3. Which area would you use in a report and why? The UTM-based area (EPSG:32643). UTM is a projected system designed to limit distortion within its zone, whereas Web Mercator is built for web display and strongly inflates area at higher latitudes. Note that Pakistan spans more than one UTM zone, and western districts such as Awaran lie outside zone 43N (72°–78°E), so their UTM distortion is larger. For a nationwide study an equal-area projection would be even better.

District	Province / Region	Calculated Area (sq km)	Rounded Area (3 Decimals)	Size Comparison
Bagh	Azad Jammu & Kashmir (AJK)	694.416529	694.417 sq km	Smallest among the five
Abbottabad	Khyber Pakhtunkhwa (KP)	1639.517137	1,639.517 sq km	~2.4x of Bagh
Astore	Gilgit-Baltistan (GB)	5132.453809	5,132.454 sq km	Moderate mountainous area
Attock	Punjab	6910.050196	6,910.050 sq km	~10x of Bagh
Awaran	Balochistan	24989.608970	24,989.609 sq km	Largest (Massive arid district)

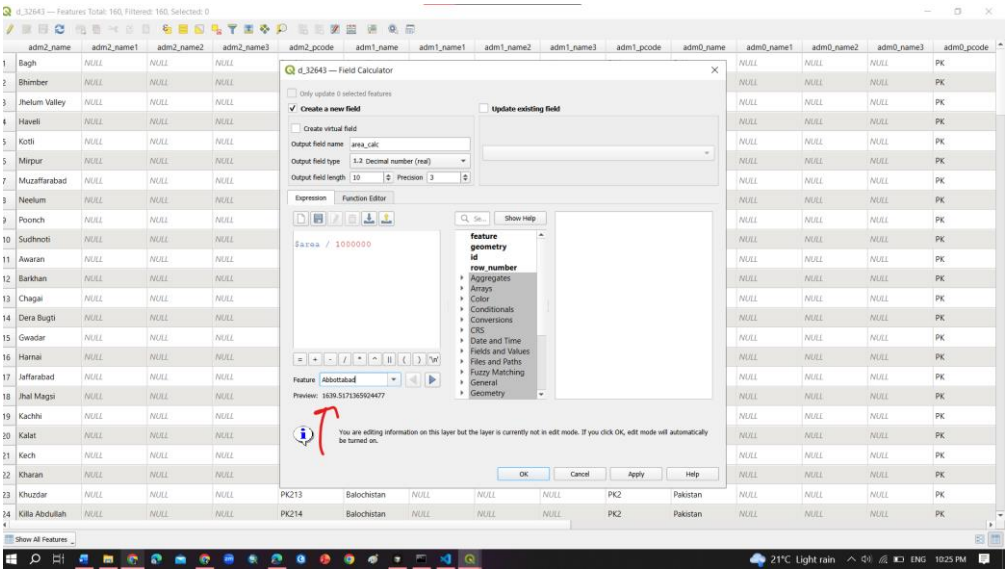
Checkpoint 9: calculated area summary for the five districts.

11.4 QGIS Field Calculator evidence

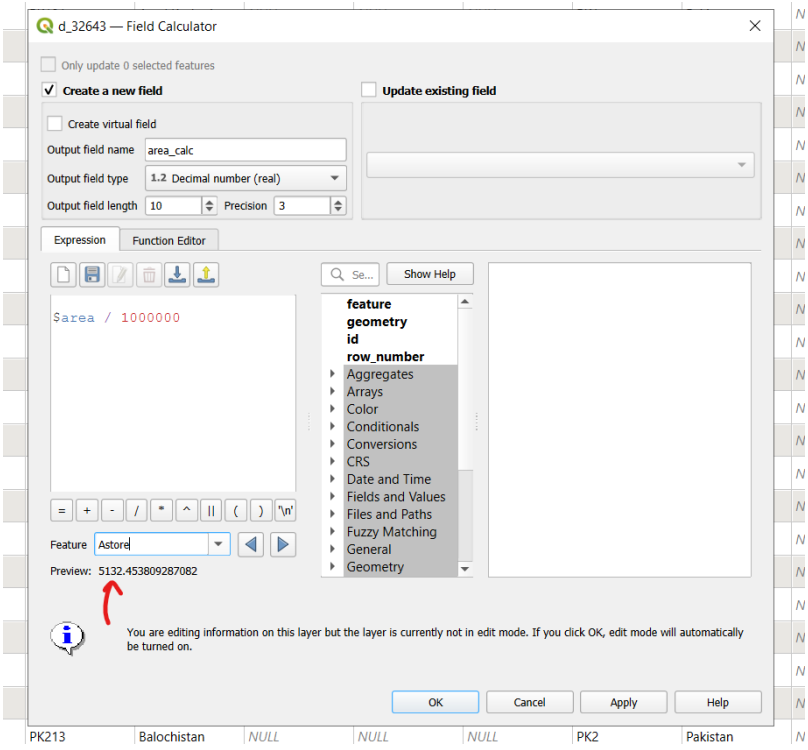
The following screenshots show the Field Calculator preview of $area / 1000000 on the d_32643 layer for each district.



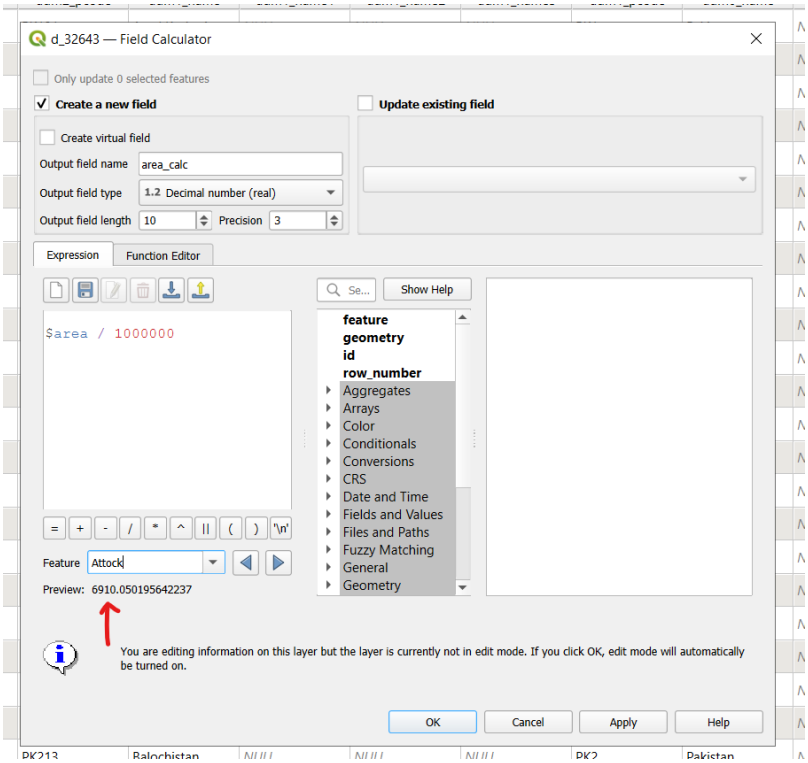
Field Calculator preview: Bagh — 694.417 km²



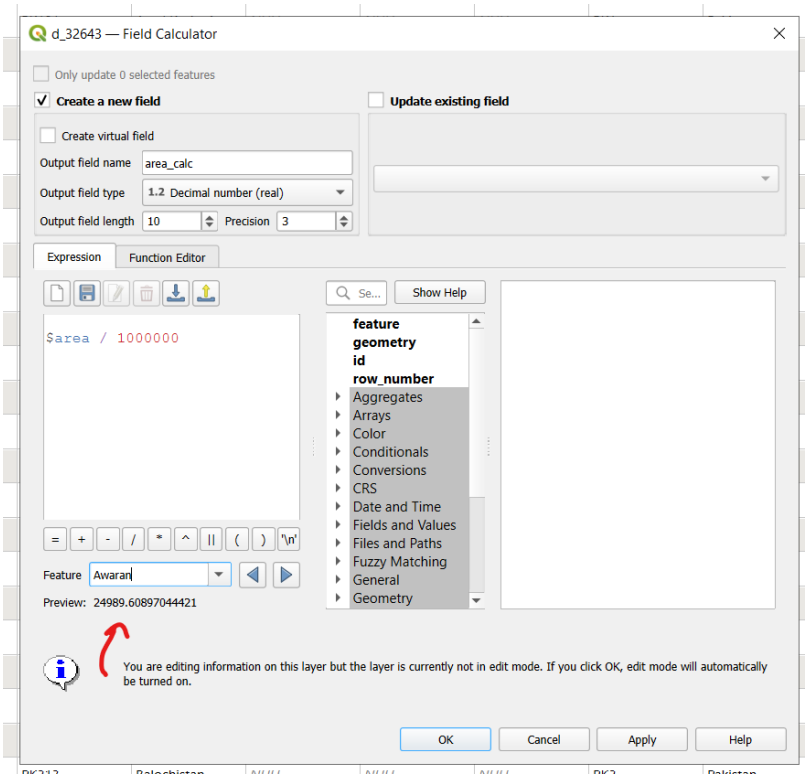
Field Calculator preview: Abbottabad — 1,639.517 km²



Field Calculator preview: Astore — 5,132.454 km²



Field Calculator preview: Attock — 6,910.050 km²



Field Calculator preview: Awaran — 24,989.609 km²

12. Written Questions

Q1. When is Version D the wrong choice?

Version D is unsuitable for detailed district-level analysis or close-up maps. The 1 km simplification removes boundary detail and can distort district shapes, as seen in the QGIS comparison with Version A.

Q2. Why is one GeoJSON request different from tile requests?

The browser requests the GeoJSON file once to obtain all district geometries and attributes. The basemap, by contrast, is delivered as many separate 256 × 256 px image tiles, requested according to the current zoom level and map extent, and new tiles are requested when the map is panned.

Q3. Why does the map container need a height?

An empty HTML div has no height by default. Without a defined height, Leaflet has no visible area in which to draw the map, so nothing appears.

Q4. What happens if draw does not remove the old layer?

Each redraw adds another district layer on top of the previous one. This creates overlapping features, makes the filter appear not to work, and can slow the map down.

Q5. Why might all districts appear in the same colour?

The population values may contain invalid text, thousands separators or placeholders such as “NA”, so the numeric comparisons fail. The code may also be reading the wrong population field. Inspecting the attribute table and feature.properties in the Console shows which of these is the cause.

Q6. Why does double-clicking index.html fail to load districts?

Double-clicking opens the page with a file:/// URL, and browsers can block JavaScript fetch requests from the local file system. Live Server serves the page over HTTP, which allows the GeoJSON request to work.

13. Conclusion

The District Explorer map loads all 160 districts, shades them by population with a legend, supports popups and province filtering, and was tested through the Console and Network tabs. The measurements show that simplifying and reducing GeoJSON precision lowers load time, and that Web Mercator areas are inflated at higher latitudes compared with UTM.